

JUBIN PHILIP^[4pt]

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Backend-focused software developer with experience building scalable backend systems, asynchronous processing pipelines, data-intensive applications, and AI-powered applications. Experienced in LLM integration, Retrieval-Augmented Generation (RAG), vector search, document processing, PostgreSQL/pgvector, and high-throughput backend systems. Experienced in leveraging Claude Code and Google Antigravity to accelerate development, debugging, and implementation.

TECHNICAL EXPERTISE

- AI LLM: RAG, LLM Integration, Embeddings, Semantic Search
- Machine Learning: PyTorch, Computer Vision, OCR, Model Training
- Frontend: React.js, Next.js, HTML5, CSS3, JavaScript, TypeScript
- Backend: Python, FastAPI, Flask, Node.js, Express.js, NestJS, GraphQL, REST APIs, Laravel
- Databases & Vector Search: PostgreSQL, pgvector, HNSW, MySQL, MongoDB
- Caching Queues: Redis, RQ, Background Workers, Asynchronous Processing
- Data Engineering: ETL Pipelines, Data Processing, Web Scraping, PDF Parsing, Document Processing
- AI Development Tools: Claude Code, Google Antigravity
- Languages: Python, JavaScript, TypeScript, PHP, Java, C

PROFESSIONAL EXPERIENCE

Junior Software Developer

Mar 2025 – Present Digilaw Legal Software Private Limited, Kochi

- Designed and implemented Redis-based queue systems for asynchronous task execution, improving system scalability and responsiveness.
- Built robust ETL pipelines to process legal data from web scraping and PDF extraction workflows, ensuring structured and reliable ingestion into PostgreSQL.
- Developed background worker services using Redis queues to handle heavy operations such as scraping, parsing, and bulk database updates.
- Decoupled synchronous APIs from long-running tasks using queue-based architecture, significantly reducing API latency and improving user experience.
- Implemented retry mechanisms, logging, and failure handling in worker pipelines to ensure fault tolerance and data consistency.
- Improved database performance by 30% through query optimization and indexing.
- Led migration from MySQL to PostgreSQL with zero data loss and improved query efficiency.
- Built secure REST APIs with JWT authentication and modular backend architecture.
- Developed and trained a computer vision model for OCR-based image text recognition using PyTorch, working with approximately 10,000 labeled images.
- Leveraged Claude Code and Google Antigravity to accelerate development, debugging, code generation, and implementation of backend and AI-related features.

Software Developer Trainee

Aug 2024 – Feb 2025 Techfriar Technologies, Infopark, Kochi

- Developed full-stack applications using MERN and PERN stacks following industry best practices.
- Engineered efficient GraphQL APIs, reducing over-fetching and improving application performance.
- Built scalable backend systems using Node.js and PostgreSQL.
- Worked in agile teams to deliver production-ready applications with proper documentation.
- Optimized application performance and improved user experience for internal tools.

PROJECTS

InkDex – AI Document Intelligence & RAG Platform

FastAPI, PostgreSQL, pgvector, Sentence Transformers, Redis, RQ

github.com/jubinphilip/InkDex-server

- Built an AI-powered document question-answering platform using Retrieval-Augmented Generation (RAG) to answer user queries from uploaded documents.
- Implemented document processing pipelines including PDF extraction, recursive text chunking, metadata handling, and vector embedding generation using Sentence Transformers.
- Integrated Gemini LLM to generate context-aware answers using semantically retrieved document chunks.

- Implemented semantic vector search using PostgreSQL and pgvector with cosine similarity, Top-K retrieval, and similarity thresholding.
- Optimized vector retrieval using HNSW indexing with cosine distance for efficient approximate nearest-neighbor search.
- Designed the backend using FastAPI with modular Router, Service, and Repository layers, SQLAlchemy, and Alembic migrations.
- Implemented Redis and RQ background workers for asynchronous document processing, chunking, and embedding generation.
- Implemented JWT authentication, authorization, and document ownership controls to securely isolate user data.

Event Management Platform

Next.js, NestJS, GraphQL, PostgreSQL

- Developed a full-stack event management web application enabling users to create, manage, and book events seamlessly.
- Built a scalable backend using NestJS and GraphQL, supporting efficient data querying and real-time interaction patterns.
- Designed PostgreSQL schemas for handling events, users, bookings, and role-based access control.
- Implemented authentication and authorization using Firebase Auth with secure API handling.
- Developed responsive frontend using Next.js with optimized rendering and state management.
- Applied modular architecture and clean code principles to ensure maintainability and scalability.

Car Rental Platform

Next.js, Node.js, Express, GraphQL, PostgreSQL

- Developed a scalable car rental system with advanced filtering, booking management, and admin dashboard.
- Implemented OTP authentication using Twilio and email verification via Nodemailer.
- Built GraphQL APIs using Apollo Server and Prisma ORM.
- Integrated MinIO object storage, improving media load performance by 35
- Designed backend logic for handling concurrent bookings and transaction safety.
- Integrated Razorpay demo payment gateway for seamless and secure booking transactions.

EDUCATION

Master of Computer Applications
CGPA: 8.31 [6pt]
Bachelor of Computer Applications
CGPA: 7.4

2022 – 2024 Nirmala College, Muvattupuzha
2019 – 2022 Don Bosco Arts and Science College, Kannur

CERTIFICATIONS

- The Complete JavaScript Course – Udemy
- Python for Data Science – NPTEL
- MERN Stack Internship – Camerin Folks